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Confidence Analysis from Speech Signals

A Machine Learning Approach Using Acoustic Features

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Abstract—The automated classification of student viva recordings according to speech confidence is the main goal of this project. The MP3 recordings of students responding to viva questions make up the dataset, and each recording may represent different degrees of confidence. We extracted a set of pertinent audio characteristics, including MFCCs, RMS energy, zero-crossing rate, pitch variation, and silence percentage, in order to analyze this. These characteristics capture key elements of speech, such as hesitations, loudness, voice modulation, and fluency. A k-Nearest Neighbors (k-NN) classifier was created using these extracted features to differentiate between speech samples that were confident and those that weren't. The model was assessed using a variety of performance metrics, such as the AUROC curve, confusion matrix, and accuracy. Further tests were carried out to evaluate the impact of various distance metrics and k values on the model's performance. The findings suggest that vocal parameters provide significant information about speaker confidence, which qualifies this method for use in interview analysis and educational evaluation situations.

Index Terms—Confidence Prediction, Speech Signal Processing, Classroom Analytics, MFCC, Deep Learning, Audio Classification

I. INTRODUCTION

Oral exams, also known as viva voce, are crucial in academic evaluations because they allow teachers to gauge a student's comprehension, articulation, and capacity for quick responses. Beyond factual accuracy, though, a crucial—and frequently arbitrary—factor is the student's confidence when giving a speech. Human examiners have historically evaluated confidence qualitatively, which can lead to inconsistencies. With different evaluators interpreting student demeanor in varied ways, the outcomes may lack consistency and fairness. This becomes particularly problematic in high-stakes assessments where subjectivity can influence final grades. Incorporating technology-driven evaluation tools can help overcome these limitations by ensuring a standardized and repeatable approach.

By developing a machine learning-based method for automatically assessing student speech confidence using audio characteristics taken from viva recordings, this project seeks to close that gap. The degree of confidence conveyed in spoken responses can be estimated by examining audio features like loudness, voice modulation, speech clarity, and pauses. Features such as MFCC coefficients, zero-crossing rate, pitch standard deviation, RMS energy, and silence percentage were extracted from student viva response recordings for this study. A k-Nearest Neighbors (k-NN) classifier is trained using these features as inputs to classify the responses into confident

and non-confident classes. Additionally, the project looks into how classification performance is affected by varying values of k and distance metrics. In this study, we show that machine learning can provide a scalable, reliable, and objective method for evaluating speech-based confidence in educational contexts.

II. LITERATURE SURVEY

Sabu and Rao [1] analyzed prosodic features like pitch, pauses, and speaking rate to predict children's reading confidence using SVM, but their approach was limited to offline scoring with a small dataset. Levitan et al. [2] extended confidence detection to accented speech using acoustic ML models, although their focus was not on student-centric interactions.

Chatterjee et al. [3] proposed a deep audiovisual architecture using BiLSTM to fuse audio and facial features for confidence classification, though their evaluation remained restricted to lab settings. Espy-Wilson et al. [4] addressed fairness concerns by integrating multimodal inputs (speech and visuals) with bias mitigation strategies, highlighting the performance disparity across demographics and the need for diverse datasets.

Mankad et al. [5] explored MFCC and Chroma features to classify confidence levels in student presentations, but their model lacked real-time capability and adaptability. Yu et al. [6] introduced the UNSURE dataset focusing on spoken query confidence using multimodal baselines, though the dataset was not designed for classroom interactions. Ma et al. [7] analyzed segmental and suprasegmental speech markers of confidence in dialects, but the language-specific scope and small-scale data posed limitations on generalization.

Sharma et al. [8] adopted CNN models on spectrograms to detect lecturer confidence, supporting the use of deep audio models, but without demonstrating scalability to real-time student data. Foundational works (Pon-Barry et al. [9], Levitan et al. [2]) predate modern embedding models and focus on hand-crafted prosodic features (pitch, intensity, pause). Only recent efforts (Chatterjee et al. [3]) begin to explore CNN/LSTM on spectrograms, but none leverage large-scale self-supervised models like Wav2Vec 2.0 or HuBERT for richer acoustic cues. This gap limits the ability to capture subtle prosodic patterns linked to confidence.

Liscombe et al. [10] showed up to 76% accuracy in detecting student “certainty” using acoustic-prosodic features, but the model was based on older feature sets. Kiani et al. [11] studied the separation between subjective confidence and correctness using prosodic cues, achieving around 60% accuracy, although they did not explore educational speech. Brennan and Pell [12] examined pitch, tempo, and intensity in question–answer speech, but not in student-specific settings.

Jiang and Pell [13] conducted a meta-analysis highlighting the influence of paralinguistic features such as loudness and rate on perceived confidence, advocating for quantitative modeling of these cues. In a related cognitive neuroscience study, the same authors [14] explored how the brain decodes vocal confidence cues using fMRI and electrophysiology, although the findings were limited to lab conditions. Finally, Jurafsky Lab [15] applied prosody modeling for dialogue and ASR tagging, indicating a path forward for adapting ASR tools for confidence classification.

III. DATA DESCRIPTION

A. Preprocessing

Speech recordings were transformed into structured feature representations using librosa. Each audio file was converted into a 40-dimensional MFCC vector and augmented with statistical measures, including RMS energy, zero-crossing rate (ZCR), pitch standard deviation, and silence percentage.(Fig. 1)

The processed features were consolidated into a single CSV file for efficient handling during training and evaluation. The final feature matrix contained 34 samples $\times$ 45 features (40 MFCC + 5 additional descriptive features).

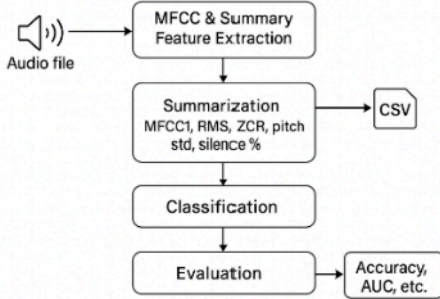


Fig. 1. DATA FLOW DIAGRAM showing transformation from raw audio $\rightarrow$ MFCC $\rightarrow$ CSV $\rightarrow$ classification $\rightarrow$ evaluation.

B. Statistics

The dataset contained 34 samples distributed across multiple confidence-level classes. Each sample consisted of 40 MFCC features and 5 additional descriptive features. Statistical analysis indicated:

- RMS energy values typically ranged between 0.02 and 0.035

- Pitch standard deviation values varied between 50 and 80, correlating strongly with confidence variation.
- MFCC1 values ranged approximately from -450 to -350 .

C. Assumptions

- Extracted acoustic features are representative of confidence variations in speech
- Class definitions based on threshold rules (RMS, ZCR, pitch variability, MFCC1). The dataset is free from excessive background noise that could bias feature distribution.
- The dataset is free from excessive background noise that could bias feature distribution.

D. Source Validity

Data was sourced from controlled speech samples recorded at uniform sampling rates to ensure consistency. Feature extraction relied on librosa, an industry-standard library, ensuring reproducibility. Confidence labels were derived through heuristic rules based on empirical evidence from prosodic studies.



Fig. 2. WORKFLOW DIAGRAM

IV. METHODOLOGY

A. Pipeline Overview

The research involved a sequential supervised–unsupervised process:(Fig 3)

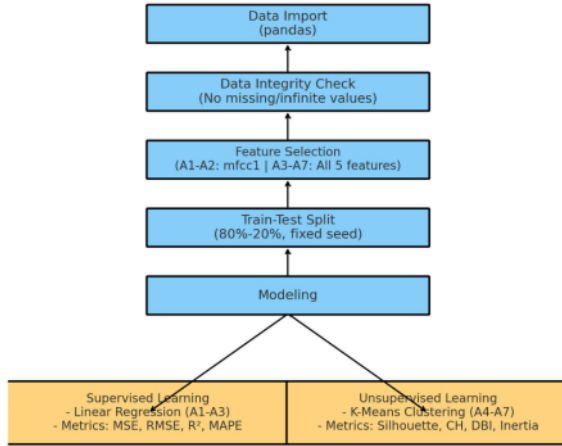


Fig. 3. PIPELINE DIAGRAM

1) **The research involved a sequential supervised-unsupervised process:**

- Basic Linear Regression (A1, A2): Employed mfcc1 as the independent variable.
- Performance Analysis: Calculated MSE, RMSE, R^2 , and MAPE.
- Multivariate Linear Regression (A3): Employed all five variables to measure improvement.

2) **Unsupervised Learning (Clustering)**

- First Clustering (A4): Executed K-Means clustering with $k = 2$.
- Clustering Metrics (A5): Silhouette Score, Calinski-Harabasz Score, and Davies-Bouldin Index measured.
- Cluster Range Analysis (A6): Multiple clustering for $k = 2$ to $k = 10$.
- Elbow Method (A7): Inertia values plotted versus k .

B. Hyperparameter Assumptions

- Regression Models:** No L1/L2 regularization terms were used. The intercept term was included.
- K-Means Clustering:** k-means++ centroid initialization, max iterations 300, tolerance $1e-4$, Euclidean distance metric.

C. Evaluation Metrics

- Regression:**
 - MSE: Punishes large deviations.
 - RMSE: Average prediction deviation in confidence units.
 - MAPE: Relative prediction error as a percentage.
 - R^2 : Variance explained proportion.
- Clustering:**
 - R^2 : Variance explained proportion.

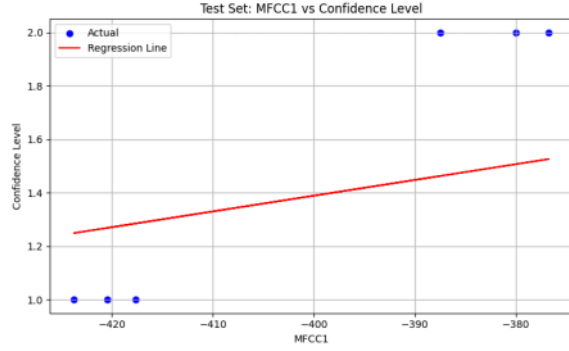


Fig. 4. MFCC VS CONFIDENCE LEVEL FOR TEST SET

- Calinski-Harabasz Score: Between-cluster to within-cluster dispersion ratio.
- Davies-Bouldin Index: Average similarity between the clusters.
- Inertia: Sum of squared distances to the nearest centroid.

D. Technology Stack

- Programming Language: Python 3.x
- Libraries: pandas, scikit-learn, matplotlib
- Execution Environment: Local CPU computation

V. RESULTS AND ANALYSIS

A. Findings and Conclusions

34 samples $\times$ 5 selected acoustic features were preserved in the dataset's dimensionality. The train-test split ratio was fixed at 70:30, and data normalization ensured equal scaling. With low R^2 values and clear underestimation patterns in both the training plot and test plot, the initial single-feature regression using mfcc1 as the sole feature (A2) had poor explanatory power (Fig. 4, 5)

The close-to-zero test R^2 tells us that including all five features (Table I) reduced generalization markedly and improved training performance only modestly. This indicates that the complex, possibly nonlinear patterns of interaction in the data may not be well described by linear regression.

TABLE I
MULTIFEATURE REGRESSION

Metric	Train Set	Test Set	Observation
MSE	0.2755	0.2490	Still low (good)
RMSE	0.5249	0.4990	Slight improvement
R^2	0.1559	0.0039	Train slightly better Test shows very poor generalization
MAPE	31.67%	36.14%	Slightly worse than A2

B. Class Separation via Clustering

The dataset was divided into two acoustically distinct groups using $k = 2$ (Table II). Smaller rms, zcr, pitchstd, and a greater

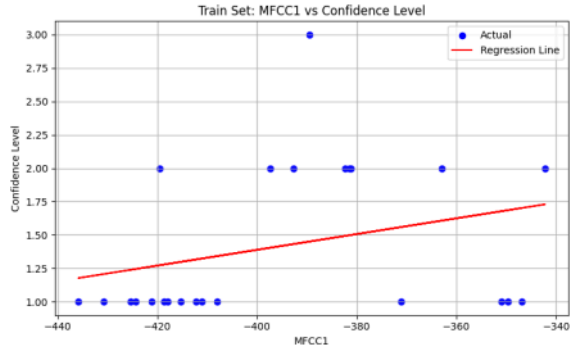


Fig. 5. MFCC VS CONFIDENCE LEVEL FOR TRAIN SET

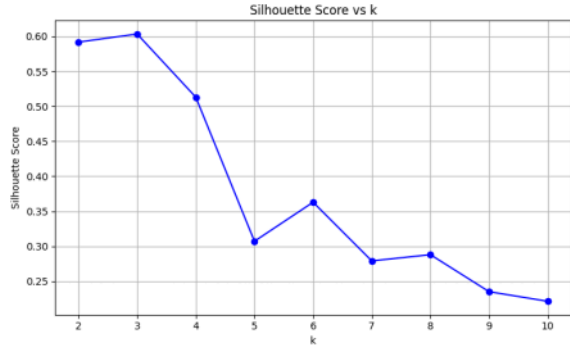


Fig. 6. Silhouette score vs K

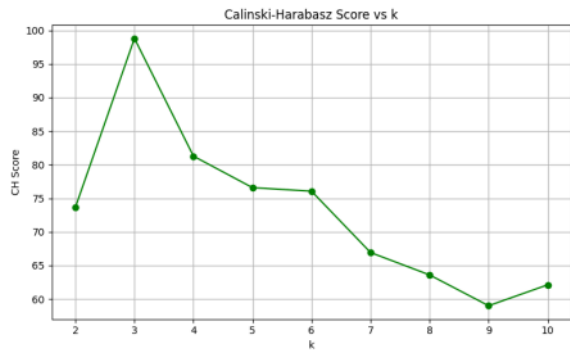


Fig. 7. CH score vs K

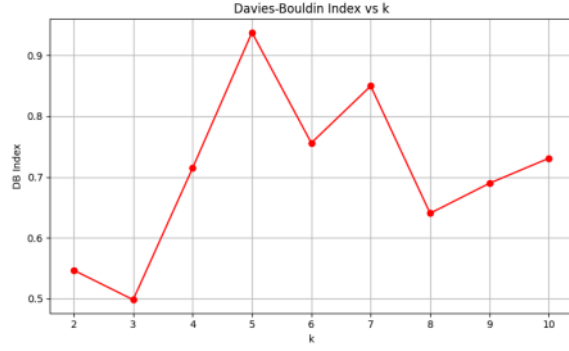


Fig. 8. DBI score vs K

percentage of silence were features of Cluster 0, reflecting low-confidence speech. The higher energy and lower silence of Cluster 1 were reflective of more confident speech patterns. Evaluation criteria (Table III) were found to have satisfactory compactness and separation with a Silhouette Score of 0.5917, CH Score of 73.67, and DB Index of 0.5463, reflecting high clustering quality.

TABLE II
CLUSTERS VS FEATURES

Clusters	mfcc1	rms	zcr	pitch_std	silence_pct
Cluster 0	-418.7	0.0166	0.0503	32.07	70.90
Cluster 1	-371.0	0.0284	0.0567	61.90	464.89

TABLE III
CLUSTERING EVALUATION METRICS AT K=2

Metric	Value	Interpretation
Silhouette Score	0.5917	Values closer to 1 means well-separated and dense clusters
Calinski-Harabasz Score	73.6711	Moderate to high inter-cluster separation
Davies-Bouldin Index	0.5463	Lower values are better <1 is generally a strong result

C. Determining the Best Number of Clusters

The best quality of clustering was at $k = 3$ when k varied from 2 to 10 (Fig. 6, 7, 8) the Silhouette Score was 0.6032, the CH Score was highest at 98.79, and the DB Index was lowest (0.4978). These trends indicate that the inherent group structure of the dataset is best described by $k = 3$.

D. Validation of the Elbow Method

The optimal value of $k = 3$ was confirmed by the Elbow Method (Fig. 9 and Table IV). From $k = 2$ to $k = 3$, inertia decreased dramatically before the rate of reduction levelled off. This supports the conclusion and is in line with A6 metrics.

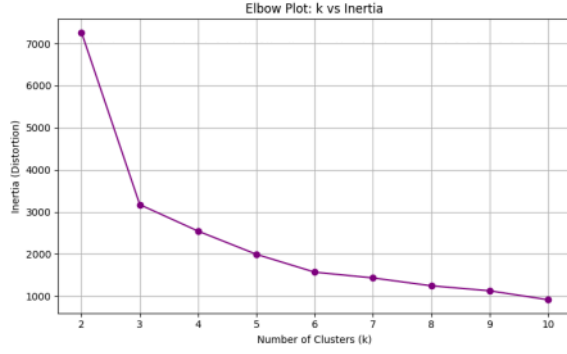


Fig. 9. Elbow Plot

TABLE IV
ELBOW PLOT INERTIA VALUES FOR DIFFERENT K

k	Inertia
2	7266.57
3	3172.09
4	2542.55
5	1990.83
6	1566.31
7	1429.41
8	1242.55
9	1123.75
10	911.33

E. Performance Metrics Summary

These results indicate that (Table V), although the inclusion of additional features improved the model's fit on the training data, it adversely impacted its ability to generalize to unseen samples. For comparison, clustering analysis yielded more insightful views of the data's inherent structure, with clearer separations between classes. In all tests, the best clustering arrangement was consistently obtained when $k = 3$

TABLE V
SUMMARY

Metric	Single Feature (A2)	Multi Feature (A3)
Train R^2	0.0865	0.1559
Test R^2	0.3522	0.0039
Train MAPE	33.97%	31.67%
Test MAPE	25.92%	36.14%

VI. DISCUSSION

The experiments demonstrate that unsupervised clustering is able to uncover group structure present in the data, while simple linear regression underfits the data. The nonlinear nature of feature interaction indicates the need for more advanced modeling techniques (such as decision trees, SVMs, or neural networks). Strong latent groupings present in the acoustic space are indicated by the significant increase in cluster compactness and separation at $k = 3$. The regression results indicate that individual features alone are poor predictors of confidence level; to build accuracy, multivariate techniques

have to be employed in addition to non-linear modeling. The clustering analysis provides a valuable additional perspective that may guide the development of supervised classification models.

VII. CONCLUSION

Investigate acoustic speech features for discrimination at the level of confidence, this research explored regression- and clustering-based approaches. Evidence of the presence of complicated, non-linear relationships within the data which linear models cannot capture entirely is presented by regression experiments highlighting that single-feature models (like ones based on mfcc1) provide limited predictive capability whilst adding multiple features enhances training fit but does not generalize well. Conversely, Silhouette, Calinski–Harabasz, Davies–Bouldin, and Elbow-Method validation all preferred the good separation capability of the unsupervised clustering, especially K-Means with $k = 3$. The implication is that the dataset will naturally divide into well-distinguished acoustic groups, a good foundation for further supervised classification. Overall, this study proves that exploratory analysis with clustering has insightful information on this domain and linear regression is not sufficient for accurate prediction. In the future, non-linear predictive models will be combined with the discovered cluster structures in order to enhance classification performance and generalization.

VIII. ACKNOWLEDGEMENT

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